

# TFPT — Red Team / Stress Test layer

Five adversarial targets (A–E): try to *break* the reductions, not confirm them

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces  $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$  as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading document 5 (Red Team).**

### TFPT status at a glance — read this card the same way in every document

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [E]/[C]       |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

The single open *structural theorem* is  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net), a constructive-AQFT theorem whose only open input is the continuum Osterwalder–Schrader reconstruction of the gapped quasi-free collar. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is now its *corollary*: a conformal net’s vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The full nonperturbative quantum-gravity measure QG.AMB.01 is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), from which TFPT is rigorously *gap-decoupled* (margin  $1.648 > 0$ , v76/v330) — an inherited, decoupled problem, *not* a TFPT-specific open item. So the honest residual is *one* solvable theorem ( $G_{\text{net}}$ ) plus the typed  $v_{\text{geo}}$  and  $F_{\text{transfer}}$  interfaces. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ .

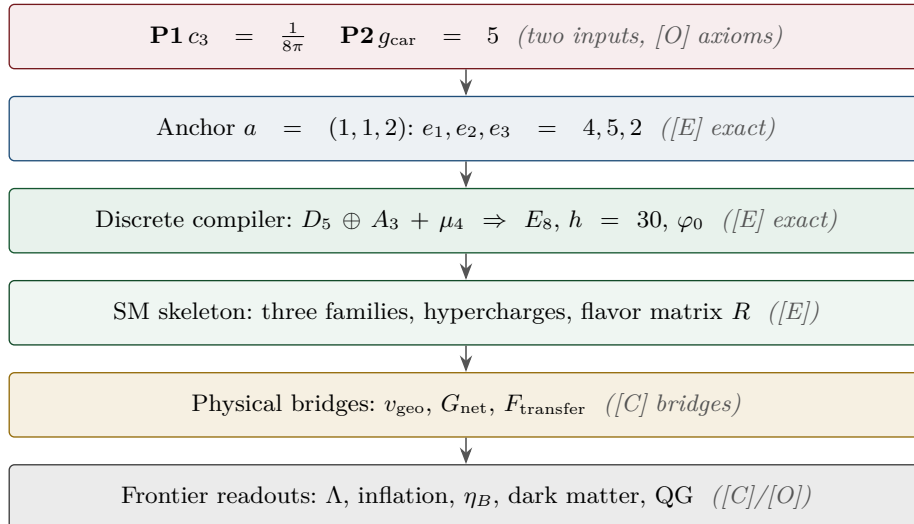
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



### What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer

is machine-backed by `[verification/redteam/rt_A_e8net.py]` ... `[verification/redteam/rt_E_vgeo.py]`, `[verification/redteam/rt_F_qft4d.py]` and `[verification/redteam/run_redteam.py]`; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass.

| Target | Minimal claim                                            | Status                     | Residual risk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|----------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A      | seam–Calderón measure = $(E8)_1$ net                     | <b>reduced, not closed</b> | <i>one</i> statement: boundary-net holomorphy + $c=8$ ( $\Leftrightarrow$ the index-4 inclusion); $E_8$ and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Since reduced to <i>zero new</i> open content: the free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the already-open $A_2$ (net assembly) + GATE.QGEO (v160–v165). The three-residual form below is the historical development. |
| B      | $g_{\text{car}} = 5$ forced by the Pascal rule           | <b>survives (narrowed)</b> | boundary derivation of half-spinor exhaustion (degree-2 truncation)                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C      | $k = c_3/2 = \frac{1}{16\pi}$ , $S/A = \frac{1}{4}$      | <b>survives</b>            | absolute $1/G$ (UV-sensitive) is the one anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| D      | ratios+products $\Rightarrow$ one scale $v_{\text{geo}}$ | <b>survives (narrowed)</b> | CP phases ( $\delta_{\text{CKM/PMNS}}$ ) not covered by $v_{\text{geo}}$                                                                                                                                                                                                                                                                                                                                                                                                                             |
| E      | one scale $v_{\text{geo}}$ carries the theory            | <b>survives (narrowed)</b> | EW/reheating/leptogenesis scales + seam=Planck identification                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Three outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), and **reduced, not closed** (the conservative wording is the correct one). A fourth, **broken**, is reserved for an actual failure; none occurred.

## Target A — the $(E8)_1$ boundary-net identification

**Minimal claim.** The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the  $(E8)_1$  lattice net, carrier subnet  $(D5)_1 \times (A3)_1$ . `[verification/redteam/rt_A_e8net.py]`

**Logical chain (established).** The Sugawara level-1 central charges are  $c(E8)_1 = \frac{248}{31} = 8$ ,  $c(D5)_1 = 5 = g_{\text{car}}$ ,  $c(A3)_1 = 3 = N_{\text{fam}}$ , so the conformal-embedding criterion  $c_{\text{coset}} = c(E8) - c(D5) - c(A3) = 0$  holds [E]. This is *compatibility*.

### Where it breaks: central charge underdetermines the net

Counterexample:  $(D8)_1 = SO(16)_1$  also has  $c = \frac{120}{15} = 8$ , but is *not* holomorphic (four primaries 1,  $v$ ,  $s$ ,  $c$ ) — a distinct  $c = 8$  net. Hence equal central charge is *necessary, not sufficient*:  $(E8)_1$  is the unique *holomorphic*  $c = 8$  net (even unimodular rank-8 lattice =  $E8$ ), so **holomorphy (single vacuum sector,  $\mu$ -index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the  $c = 8$  Narain

family is positive-dimensional. The constructive map seam–Calderón kernel  $\rightarrow (E8)_1$  and the uniqueness of bulk reconstruction are not established; the gap  $\Delta_{\text{eff}} > 0$  *supports* tightness (clustering) but does not prove it.

**Verdict.** Keep “reduced to a rigorous boundary-net identification problem”; do *not* write “metric sector closed”. The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

## Target B — carrier rank / the Pascal condition

**Minimal formal claim.**  $2^g = g^2 + g + 2$  has the unique solution  $g = 5$  (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion  $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ? [verification/redteam/rt\_B\_pascal.py]

**Where it narrows: the rule, not the arithmetic, selects 5**

Perturbing the constant breaks it:  $2^g = g^2 + g + c$  gives  $\{5\}$  only at  $c = 2$ ; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with  $2^{g-1} = \sum_{k \leq K} \binom{g}{k}$  one finds  $K=0 \rightarrow g=1$ ,  $K=1 \rightarrow 3$ ,  $K=2 \rightarrow 5$ ,  $K=3 \rightarrow 7$ , i.e.  $g = 2K + 1$ . So choosing  $K = 2$  (to reach  $g_{\text{car}} = 5$ ) is a physical postulate. It is corroborated — the  $D5$  half-spinor  $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$  and the family closure  $N_{\text{fam}} = (2^{g-1} - 1)/g$ ,  $g + N_{\text{fam}} = 8$  both pick 5 — but each is itself a postulate, not a theorem.

**Verdict.** Theorem A’s arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

## Target C — $k = c_3/2$ and the seam-area coefficient

**Minimal claim.** In reduced units  $k = c_3/2 = \frac{1}{16\pi}$  and Fursaev–Solodukhin gives  $S/A = \frac{1}{4}$ . [verification/redteam/rt\_C\_kc3.py]

**The dimensional firewall holds**

The only legal chain is

$$k_{\text{red}} = \frac{c_3}{2} = \frac{1}{16\pi} \text{ (dimensionless)}, \quad k_{\text{phys}} = \frac{k_{\text{red}}}{G}, \quad S = 4\pi k_{\text{phys}} A_{\text{phys}} = \frac{A_{\text{phys}}}{4G}.$$

The forbidden form  $k_{\text{phys}} = c_3/2$  is dimensionally false (legal only at  $G = 1$ ). A scan of v67/v68/v73 finds every  $c_3/2$  used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall’s teeth are verified against a synthetic violation. Internally consistent with  $1/G$  being UV-sensitive (Sakharov/Connes, v68).

**Verdict.** Safe as worded (reduced coefficient). Status [E] (structure) + [O] for the residual. Residual: the absolute  $4D$  Newton scale  $1/G$  ( $\propto \Lambda^2 f_2$ ) stays the one dimensionful anchor; nothing *dimensionless* is open.

## Target D — $U_{\text{point}} \rightarrow v_{\text{geo}}$

**Minimal claim.** Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale  $v_{\text{geo}}$ . [verification/redteam/rt\_D\_upoint.py]

**Where it narrows: the bijection needs explicit hypotheses**

For positive reals, (ratios, product)  $\Leftrightarrow$  (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational  $c$ , odd size  $3 = N_{\text{fam}}$ ). But it *fails* off-assumption: even sector size leaves a sign ambiguity ( $\{2, 3\}$  vs  $\{-2, -3\}$  share ratio  $3/2$  and product 6); complex amplitudes leave an  $n$ -th-root branch ambiguity ( $\{1, 1, 1\}$ ,  $\{\omega, \omega, \omega\}$ ,  $\{\omega^2, \omega^2, \omega^2\}$  share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases  $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$  are complex and

are *not* folded into  $v_{\text{geo}}$ .

**Verdict.** Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in v75 and must be named. Status [E] (reduction) + [O] ( $v_{\text{geo}}$ ). Residual: the CP-phase sector.

## Target E — $v_{\text{geo}}$ as the unique dimensional floor

**Minimal claim.** No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit  $v_{\text{geo}}$ . [verification/redteam/rt\_E\_vgeo.py]

### Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)  $\times v_{\text{geo}}$ :  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ ,  $f_a = (c_3/|\mu_4|)^{7/2} \bar{M}_{\text{Pl}}$ , masses  $= (\pi/\sqrt{2}) c_f \varphi_0^k v_{\text{geo}}$ , and  $1/G$  shares the same anchor (v68/v75). The audit *surfaces* candidates that are not pure reductions: the seam cutoff  $\Lambda$  (only an *identification* seam=Planck collapses it to  $v_{\text{geo}}$ ), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [C]/[O] inputs that must not be silently absorbed into  $v_{\text{geo}}$ .

**Verdict.**  $v_{\text{geo}}$  is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [E] (ratios) + [O] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

## Target F — the perturbative 4D-QFT + scale round (v269–v275)

**Minimal claim.** The round published in release 5.2 — the Epstein–Glaser  $S_{\text{pert}}$  layer (v269/v271/v273), the PMNS Jarlskog CP strength (v270), the absolute  $\nu$ -scale typing (v272), the scale over-determination (v274) and the QG.AMB.01 roadmap (v275). [verification/redteam/rt\_F\_qft4d.py]

### Two attacks land; the round survives once they are named

(1) **The gravity  $S_{\text{pert}}$  is non-unitary.** Power-counting renormalizability is *not* unitarity: the  $R^2/\text{Weyl}^2$  sector gives a four-derivative propagator  $1/(p^2(p^2 + M^2))$  with a negative-residue massive spin-2 mode (the Stelle ghost). So  $S_{\text{pert}}$  is unitary for the matter+gauge sector only; the gravity perturbative S-matrix carries the ghost — which is exactly why the *nonperturbative* QG.AMB.01 is a *decoupled general* QG problem (the Euclidean-QG conformal-factor problem; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), not a new gap. (2) **The over-determination is conditional.** v274's Route-2 inverts the  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  prediction, which is itself LAMBDA.BRANCH.01 [C]; so the 0.11% gravity-vs-cosmology agreement is conditional on the  $\Lambda$ -branch (the gravity route is unconditional). **Already-honest:**  $J_{\text{PMNS}}$  inherits  $\delta$ 's [C] provenance (assembled  $\mu_6$  phase, not an  $M_\nu$  output) and the  $\nu$ -scale is [O] with the NO floor  $\Sigma m_\nu = 0.0586$  eV as the one kill test.

**Verdict.** SURVIVES (NARROWED): the round stands as typed; the two narrowings above are now folded into v269 and v274. Residual: the perturbative gravity S-matrix is non-unitary (the Stelle ghost) — this is the *decoupled general* QG.AMB.01 problem (gap-decoupled from every TFPT readout, not a TFPT structural item, v335); and the anchor over-determination is conditional on the  $\Lambda$ -branch.

**Under examination: an infinite-derivative narrowing ([C], does *not* close the gate).** The Stelle ghost is a feature of the *local*  $R+R^2/\text{Weyl}^2$  *truncation*: the four-derivative propagator  $1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]$  carries the negative-residue spin-2 mode. But that truncation is the  $a_4$  Seeley–DeWitt order of the spectral action  $\text{Tr } f(D^2/\Lambda^2)$ , whose cutoff is *not* a free scheme — it is the  $\beta=1$  seam KMS weight  $f(u) = e^{-u}$  (v259, no new parameter). Keeping that exponential cutoff *un-truncated* points to a nonlocal, infinite-derivative graviton form factor  $p^2 a(p^2)$ ; for an *entire* factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator

$1/(p^2a)$  keeps its *only* pole at  $p^2=0$  — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006) — whereas any *polynomial* (finite-derivative) truncation has a real zero = the ghost pole. So *if* the resummed form factor is entire, the perturbative ghost is a truncation artefact: a conditional [C] narrowing of this attack. It does *not* close QG.AMB.01 — the trace regulator  $f$  is not yet shown to resum to an entire kinetic form factor  $a(\square)$ , and a ghost-free *perturbative* graviton is still not the nonperturbative ambient measure. (QGAMB.IDG.01, [C]/[O]; consistent with the SEAM.EQUIV.01 firewall.) [verification/v304\_idg\_nonlocal\_ghost.py]

## Outcome

**What the red team changed.** No target broke. The hardest gate (A) is confirmed *open* with the conservative wording; four targets (B, D, E, F) *survive once a silent assumption is named*; one (C) survives as worded. The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

**Follow-up — the verdicts were then used as a worklist [E]**  
([verification/v83\_e8net\_holomorphic\_uniqueness.py])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows GATE.METRIC.04 and CAR.PASCAL.01.

**Target A, residual 1 — closed [E].** At level 1 the number of primaries equals  $\det(\text{Cartan}) = |\text{fundamental group}|$ :  $A_3=4$ ,  $D_5=4$ ,  $D_8=4$ ,  $\mathbf{E}_8=1$ . So holomorphy (single vacuum sector,  $\mu$ -index 1) is *necessary* — it excludes the same- $c$  rival  $(D_8)_1 = SO(16)_1$  ( $c = 120/15 = 8$  but four primaries  $1, v, s, c$ ) — *and sufficient*: a holomorphic  $c = 8$  chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one,  $E_8$ , by the Minkowski–Siegel mass formula (mass =  $1/|W(E_8)| = 1/696729600$ , and  $|\text{Aut}(E_8)|$  saturates it). The  $(D_5)_1 \times (A_3)_1$  embedding  $c = 5 + 3 = 8$  stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with  $c = 8$  (then  $E_8$  is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy +  $c = 8$  ( $\Delta_{\text{eff}} > 0$  supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [C]/[O].

**Target B — reduced [E].** The “ $K = 2$  Pascal truncation” is not free:  $K = (g - 1)/2$  is the Pascal-row midpoint  $\sum_{k \leq (g-1)/2} \binom{g}{k} = 2^{g-1}$  (odd  $g$ ), i.e. the half-spinor split; the carrier is the even Clifford half-spinor  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$ . So B’s residual reduces to the single standard input “carrier = half-spinor of  $\text{Spin}(10)$ ” (the Lean arithmetic core stays [E]).

**Target B — a backward certificate ([verification/v219\_icosahedral\_mckay.py]).** The choice  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$  that B attacks is, *given* the  $E_8$  closure, the icosahedron: by the McKay correspondence  $E_8$  is the exceptional top of the finite- $SU(2)$  tower ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ), the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  equal to the affine- $E_8$  Kac marks ( $\sum = 30 = h(E_8)$ ), and it contains the 2, 3, 5 rotation-axis orders. This does *not* break B’s verdict — it is a *backward* certificate (it presupposes  $E_8$ ), so  $g_{\text{car}} = 5$  stays the P2 interface [O]; what it removes is the impression that  $(2, 3, 5)$  is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Still open, typed honestly: A (i)+(ii); Target D’s CP-phase residual (at  $\theta_{13} = 0$  the Dirac phase is undefined, so this needs a texture correction); Target E’s reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

**Second follow-up round [E]**  
([verification/v87\_bulk\_uniqueness\_reduction.py],  
[verification/v88\_cp\_phase\_audit.py], [verification/v86\_nstar\_reheating.py])

**Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual.** Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems  $\simeq$  physical modular invariants of  $\text{Rep}(A)$ . For a *holomorphic* net  $\text{Rep}(A) = \text{Vect}$ , so the bulk pairing is *unique*. The contrast is machine-checked: the same- $c$  rival  $SO(16)_1$  (discriminant category  $\mathbb{Z}_2 \times \mathbb{Z}_2$ ,  $\mu$ -index 4) admits *six*



modular invariants — diagonal,  $s \leftrightarrow c$  swap, both  $E_8$ -extension invariants  $|\chi_0 + \chi_{s/c}|^2$  (the lattice glue  $E_8 = D_8 \cup (D_8 + s)$ ,  $h_s=1$  bosonic), and two twisted pairings — so without holomorphy the  $c=8$  bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem “seam–Calderón boundary net is holomorphic with  $c=8$ ” ([C]/[O]; ledger GATE.METRIC.05). The theorem also has an equivalent, more tractable *index* form ([verification/v89\_carrier\_index\_lemma.py], ledger GATE.METRIC.06): by Kawahigashi–Longo–Müger,  $\mu_A = [B:A]^2 \mu_B$ , the carrier inclusion has Jones index  $[(E_8)_1 : (D_5)_1 \times (A_3)_1] = \sqrt{16} = 4 = |\mu_4|$  — the glue-group order *is* the inclusion index — and the extension is the  $\mu_4$  simple-current extension (all three glue sectors are  $h=1$  currents;  $248 = 45+15+64+64+60$ ). Since an index-4 simple-current extension of the carrier has  $\mu = 16/4^2 = 1$ , **holomorphy follows from the index statement**: Target A  $\Leftrightarrow$  “the seam net contains the carrier net with Jones index 4 as its  $\mu_4$  simple-current extension” — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([verification/v92\_glue\_uniqueness.py], ledger GATE.METRIC.07): exhaustive classification of the carrier discriminant form  $(\mathbb{Z}_4 \times \mathbb{Z}_4, q=(5x^2+3y^2)/8)$  shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet  $\mathbb{Z}_2$ ; the same two objects as the  $E_8$ -extension invariants above), and exactly one halfway extension, whose induced form  $q = \{0, 0, 0, \frac{1}{2}\}$  is the  $D_8$  discriminant form: the  $SO(16)_1$  rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier ( $\mu=16$ )  $\rightarrow SO(16)_1$  ( $\mu=4$ )  $\rightarrow (E_8)_1$  ( $\mu=1$ , sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

**Target A, final form — the Simple-Current Extension Theorem** [E]/[C] ([verification/v154\_simple\_current\_theorem.py]). Assembled into one statement: the carrier net  $A = (D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L = \langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$ . The *entire* adversarial residual of Target A is now the single seam-side identification “the RP seam–Calderón net *is*  $A$ ” — the algebra  $(B \cong (E_8)_1)$  is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core (“the kernel is quasi-free”) *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN =  $|k|$ , 16 free Majoranas, currents  $248=120+128$ , holomorphic character  $E_4/\eta^8 = j^{1/3}$  with 248 at level 1 (v156). *Final reduction* (v160–v165): the free-bulk premise is recast as a fixed-point theorem (quasi-free  $\Rightarrow$  all higher cumulants  $\kappa_{2n}=0$ , v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap  $(2/3)^6$  (v161), whose value is forced by  $\mu_4$ -equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items —  $A_2$  net existence (an assembly of established net constructions, [C]) and  $R_1 = \text{GATE.QGEO}$  (“seam boundary =  $\mathbb{P}^1 \setminus \mu_4$ ”) — with the irreducible core  $\{\pi, v_{\text{geo}}\}$  a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. The two structural residuals are then discharged to a theorem (v175): the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  makes full-cone reflection positivity reduce *for every*  $m$  to the one-particle contraction (verified on the complete  $2^{16}$ -dim Fock space), and  $A_2$  is an assembled, verified  $(E_8)_1$  certificate; net existence and full-cone RP become [E]. Target A’s single residual is therefore the seam realisation, now the named keystone SEAM.EQUIV.01 — *the raw RP seam is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — whose conformal-deck premise QGEO.SYM.01 is its *downstream* definitional face, Lean-pinned (FORM.SEAMEQUIV.01, v308) to named cited steps plus the derived Recovery gap. The sprint-by-sprint reduction (v176–v302) is recorded in the changelog, not replayed here.

**Target D — the CP-phase residual quantified** [E] (not closed). With the frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$ : pull vs  $\gamma_{\text{PDG}} = 65.7^\circ \pm 3.0^\circ$  is  $+0.98\sigma$  (survives). Audit, *not* promoted: the data central value sits  $0.07^\circ$  from the alternative coefficient reading  $\pi/3 + 2\lambda_C^2$  ( $|\mathbb{Z}_2|$  instead of  $N_{\text{fam}}$ ) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (REG.FREEZE.01). Decision threshold: the readings differ by  $\lambda_C^2 = 2.88^\circ$ , i.e.  $3\sigma$ -distinguishable once  $\sigma_\gamma \leq 0.96^\circ$ . The  $J$ -inversion is flagged magnitude-contaminated (source-scale  $s_{23}$  vs  $M_Z$ ). Ledger FLAV.CP.01.

**Target E — one flagged scale computed** [C]. The reheating temperature is no longer a silent input:  $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  [E] + standard physics give  $T_{\text{reh}} = 9.6 \times 10^9$  GeV and  $N_*(0.05/\text{Mpc}) = 51.4$  [C] (ledger COSMO.NSTAR.01); honestly recorded: this slow Higgs-channel point is  $A_s$ -disfavoured ( $-11.4\sigma$ ; the measured  $A_s$  requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band [50, 60] stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was merged (A), one was quantified (D), one input was computed (E).**

**Target D — the CP residual geometrically *located* (not closed) [E]/[C]**

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. **(i) A channel typing** ([`verification/v227_degree_exponent_channel_split.py`]). The  $E_8$  split  $248 = 120 + 128$  is a magnitude/phase split: the exponents sum to  $120 = |R^+(E_8)|$  (the *magnitude* channel, where the (ratios, product)  $\Rightarrow v_{\text{geo}}$  bijection lives), the fundamental degrees sum to  $128 = \text{rank } E_8 \cdot \dim S^+$  (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong “ $S^-$  is dark matter” reading). **(ii) A hexagonal modulus** ([`verification/v220_cp_hexagonal_modulus.py`]). The seam deck is the *square* modulus ( $j=1728, \mathbb{Z}/4$  clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus  $j=0$  ( $\tau=\rho=e^{i\pi/3}, \mathbb{Z}/6$ , CM by  $\mathbb{Z}[\omega]$ ). The frozen leading reading  $\delta = \pi/3 + 3\lambda_C^2 = 68.65^\circ$  has leading term  $\pi/3 = \arg \rho$ , the hexagonal CM phase; CP lives in the  $\mathbb{Z}/6$  phase *fiber* over the  $\mathbb{Z}/4$  seam, not in the deck. **(iii) An orientation** ([`verification/v225_dual_normal_frame.py`]). The dual normal frame  $d = a^\top R^{-1} = -\frac{1}{2}(1, 1, -2)$  (Nariai/horizon normal) and  $n = (5, -9, 6)$  (first-generation selector,  $n^\top R = (8, 0, 0)$ ) has oriented volume  $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}} \cdot 7$ , and rotating by the hexagonal unit gives  $\text{Im} \det(\mathbf{1}, d, \rho n) = 21 \sin(\pi/3) \neq 0$  — CP sits precisely in the *orientation* of the normal frame, exactly where the magnitude bijection fails. All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry  $\delta$  is untouched). **(iv) One hexagonal unit, two sheets** ([`verification/v231_cp_mu6_phases.py`]). The reduction is sharper still: *both* CP phases are  $\mu_6$  powers of the single hexagonal CM unit  $\rho = e^{i\pi/3}$  —  $\delta_{\text{CKM}}^{\text{lead}} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the neutrino phase lattice) — with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  is the quark phase times the  $\mathbb{Z}_2$  sheet ( $\rho^3 = -1$ ); the frame orientations of (iii) are correspondingly sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). So Target D’s CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual  $3\lambda_C^2$  (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. **(v) The universal triality phase** ([`verification/v233_cp_triality_phase.py`]). In the canonical  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation,  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight); since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of  $\mu_6$  power: the power choice of (iv) is removed, and  $\omega$  is the  $\mu_3$  factor of the order-30 (2, 3, 5) Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C]. **(vi) The lock as a pre-registered kill test** ([`verification/v320_galois_cp_relation.py`]). Because  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  is now *forced* (one unit  $\rho = \zeta_6$ , the  $\mu_4$ -power  $\rho^4$ , the  $\mathbb{Z}_2$  sheet  $\rho^3 = -1$ ), it is a falsifiable cross-prediction, not a free input: a measured  $\delta_{\text{PMNS}}$  robustly away from  $240^\circ$  at DUNE/Hyper-K (beyond the sub-leading budget,  $> 3\sigma$ ) kills the Galois-CP organisation. The relation is [E], the physical phase reading stays [C], the kill test is [X].

**Global look-elsewhere closure [E] ([`verification/v100_numerology_null_mc.py`])**

The per-target look-elsewhere traps recorded above (the  $\pi/3 + 2\lambda_C^2$  alternative in Target D, the  $\ln(46080)$  scale reading destroyed in [`verification/v99_koide_flow_time.py`]) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability  $\prod_i p_i = 10^{-25.8}$  over the 13 scored observables; all 94 500  $F_{U(1)}$  variants solved, exactly one hits CODATA ( $p_\alpha \leq 1.1 \cdot 10^{-5}$ ); total  $\leq 10^{-30.7}$ . The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT’s own hit count  $13 \rightarrow 9/6$ ; data shuffling  $\rightarrow 1.2$ ) prove it has power, and the census is stable under budget/window variation ( $< 2$  orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (tfpt\_3) and the module docstring.



| Structural firewall + out-of-sample seed cross-validation [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| ([verification/v305_witness_independence.py], [verification/v306_seed_crossval.py])                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>The global null test answers “is the <i>scorecard</i> fishing?”; two companions answer the deeper adversarial worries “is the same integer counted many times as if independent?” and “is one shared seed dressed up as many predictions?”. <b>Witness economy</b> ([verification/v305_witness_independence.py]): every headline integer (<math> \mu_4 =4</math>, <math>g_{\text{car}}=5</math>, <math> \mathbb{Z}_2 =2</math>, <math>N_{\text{fam}}=3</math>, <math>\dim S^+=16</math>, <math>\text{rank } E_8=8</math>, 240, 248, <math> 2I =120</math>, <math>h=30</math>, <math> W(D_5) =1920</math>, <math>\Omega_{\text{adm}}=48</math>, the gap exponent 6) is a documented image of the <i>single</i> anchor <math>a=(1,1,2)</math> with <math>\pi</math> the only transcendental primitive, so the recurring <math>2/3= \mathbb{Z}_2 /N_{\text{fam}}</math> and <math>(2/3)^6</math> are <i>one</i> ratio, not many — 13 atoms from 2 free primitives (<math>6.5\times</math> compression made explicit). The <math>E_8</math> spine is genuinely overdetermined (<math>\geq 3</math> structurally distinct derivations sharing only leaf inputs), while the pure-seed readouts are single-witness, so their joint surprise is scored jointly by <b>v100</b>, never multiplied; the negative control shows <math>\alpha^{-1}</math>’s “137” is <i>foreign</i> to the anchor family (an independent witness via the <math>F_{U(1)}</math> cubic, <b>v3</b>). <b>Seed cross-validation</b> ([verification/v306_seed_crossval.py]): treating the axiom seed <math>\varphi_0</math> adversarially as a free dial and back-solving it from each of the five seed-grammar observables gives an error-weighted mean reproducing <math>\varphi_0 = 1/(6\pi) + 48c_3^4</math> to 0.04%; leave-one-out predicts every held-out observable within <math>3\sigma</math> <i>out of sample</i> (the most-tensioned hold-out is <math>\sin^2 \theta_{13}</math>, <math>\sim 2\sigma</math>), and a data<math>\leftrightarrow</math>formula shuffle explodes the cross-validation <math>\chi^2</math> by <math>&gt; 100\times</math> (the test has power). Both stay [E] as structural/statistical checks, conditional on the same declared grammar. <b>No-golden-dynamics rule</b> ([verification/v305_witness_independence.py]): the golden ratio <math>\phi = 2\cos(\pi/5)</math> lives in the <i>static</i> carrier field <math>\mathbb{Q}(\sqrt{5})</math> (the <math>\mathbb{Z}/5</math> hand; minimal polynomial <math>x^2 - x - 1</math>, discriminant 5 non-square), while every <i>dynamic</i> rate is rational (<math>\mathbb{Q}</math>, the <math>\mathbb{Z}/6</math> = sheet<math>\times</math>family hand, e.g. the recovery factor <math>(2/3)^6</math>) — so a dynamic rate “explained” by <math>\phi</math> is a cross-field false friend, and CP phases and recovery rates must run on the order-6 hand (the field firewall of <b>v314/v315</b>).</p> |

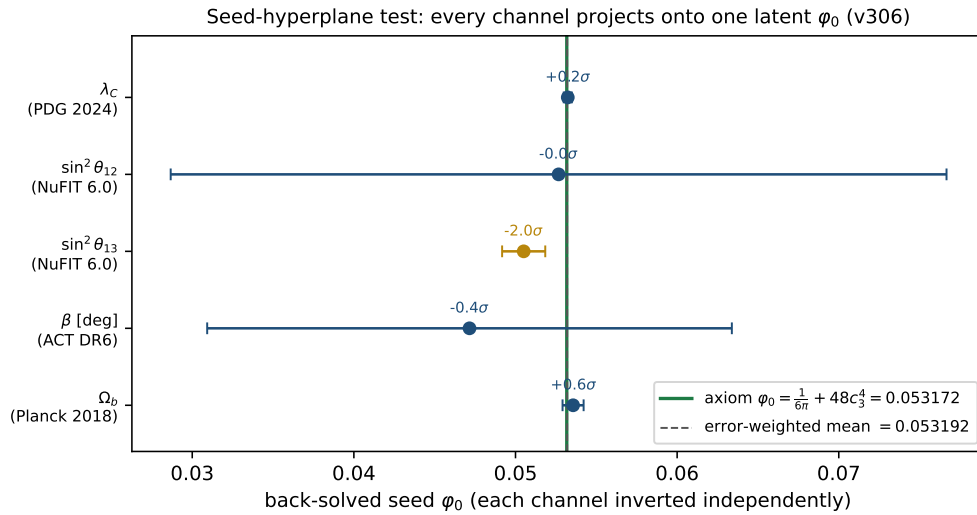


Figure 1: The seed-hyperplane test ([verification/v306\_seed\_crossval.py]): each scorecard observable is *independently* inverted to the latent seed  $\varphi_0$  (with its measurement error). All five project onto the one axiom value  $\varphi_0 = \frac{1}{6\pi} + 48c_3^4 = 0.053172$  (green), and their error-weighted mean reproduces it to 0.04% — one real seed, not five fits. The honest pressure point is  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  low, a *separate* carrier-trace channel, **v268**); a data $\leftrightarrow$ formula shuffle explodes the cross-validation  $\chi^2$  by  $> 100\times$ . [E].

| External-review residual map [E]/[C] ([verification/v182_reviewer_residual_map.py])                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: <b>(A)</b> the mapping <math>2D \rightarrow 4D</math> (“why does the dynamics read the masses as the Plücker norm 11?”), <b>(B)</b> the abstractness of UV gravity, <b>(C)</b> the meta-look-elsewhere of the grammar choice, and <b>(D)</b> the Koide Möbius flow (non-integer flow time <math>t \sim 2.84</math>, missing QFT generator). None adds an <i>independent</i> open gap. <b>A</b> [E]/[C]: the readout integer is exact (<math>11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}}</math>, the QBL boundary count, <b>v174</b>) and the <math>2D \rightarrow 4D</math> step</p> |

is the holographic reduction, so  $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . **B [E]**: “gravity is the geometric readout of the boundary” *is* **QGEO.SYM.01**. **D [E]/[C]**:  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  is the Möbius group and the cusp flow lives on the deck  $\mathbb{P} \setminus \mu_4$ , so the Möbius *nature* is forced by the geometry; the non-integer  $t \sim 2.84$  only locates the physical point and the missing generator *is*  $F_{\text{transfer}}$ , so  $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (**v84**), minimal and over-determined (the anchor  $a = (1, 1, 2)$  lands in  $\geq 6$  mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay **[O]/[C]**.